

Characterising measurements of vessel diameter in optical coherence tomography angiography and correcting artefact-induced underestimation

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Abstract:

Optical coherence tomography angiography (OCTA) metrics are widely used to assess the retinal vasculature, and vessel diameter has been linked to a range of vascular diseases. However, shear-induced alignment of red blood cells at vessel edges reduces backscattering and suppresses the OCTA signal, which can lead to systematic underestimation of vessel diameter. In this study, we established ground-truth diameters in the in vivo mouse retina using Intralipid contrast enhancement, compared conventional binary-mask-based calculation against model-based profile fitting, and used flicker stimulation to evaluate the performance of each method. We additionally investigated these methods on maximum and mean intensity projection data to determine the best configuration. Before contrast enhancement, a generalised Gaussian fit to the maximum intensity projection underestimated diameter by approximately 35%, but retained the strongest linear relationship with ground truth of all methods tested, making it correctable by calibration. Post-contrast, the same fit applied to the mean intensity projection was the most reliable in our evaluation. In a test of flicker stimulation induced vasodilation, model-based fitting detected the expected vasodilation, whereas binary-mask-based calculation was more affected by intensity fluctuations and instead reported vasoconstriction. These findings characterise OCTA diameter underestimation and establish model-based fitting as a practical means of correcting it.

1. Introduction

Different retinal diseases have been shown to preferentially affect different vascular plexuses, with glaucoma altering the superficial vascular plexus [1] while diabetic retinopathy [2] and retinal vein occlusion [3] affect the deeper plexuses more severely. These vascular changes are commonly quantified using vessel density [4–7], which is a measure of the proportion of the plexus occupied by visible vessels in the projected angiogram [8, 9]. However, certain density measurements such as vessel area density are influenced not only by the number and distribution of vessels, but also by the vessel diameter since the projected area of a vessel scales with its measured calibre [8, 10]. Vessel diameter is therefore an important biomarker, providing information about local and systemic vascular health. Thickening of the arteriolar wall has been shown to narrow the vessel lumen in proliferative diabetic retinopathy [11]. Retinal arteriolar narrowing has been shown to precede and predict the onset of systemic hypertension [12]. In glaucoma suspects, reduced vessel diameter has been shown to track both retinal ganglion cell dysfunction and visual field loss [13]. Diameter changes are also informative beyond the eye, where a meta-analysis of observational studies found smaller retinal arteriolar calibre to be associated with coronary heart disease [14]. Combined with measurements of blood flow, diameter can also be used to investigate local blood pressure [15, 16]. Therefore, it is imperative that accurate measurements

of vessel diameter be used when making these extrapolations or when modelling behaviours of flow in digital twins [17, 18]. Optical imaging techniques are often used to measure vessel diameter due to their high resolution, but accurate measurement of vessel size is more difficult than it first appears.

Optical coherence tomography (OCT) is a non-invasive optical imaging method, which uses interferometry to measure the transit time of light that is backscattered from a sample, thereby producing depth-resolved images of tissue [19, 20]. Optical coherence tomography angiography (OCTA) is an extension of OCT that visualises vascular perfusion using the motion contrast from red blood cells (RBCs) [21–24]. Over the years, OCTA has demonstrated widespread acceptance and use for clinical evaluations of vascular features in relation to disease [25]. OCT angiograms can be used to evaluate characteristics on the scale of vascular layers by measuring vascular density or non-perfusion zones [1–3, 26–28], or it can be used to gauge vessel size [29] or tortuosity [30].

Analysis of OCTA datasets is typically achieved by performing a maximum intensity projection of a 3D vascular network to compress it into 2D before binarising the resulting angiogram using a global or local binarisation method [8, 9]. From this binarised angiogram, various vascular metrics can then be quantified. Importantly, there are many methods for performing this binarisation step and the choice of method can significantly affect the reported vascular metrics, with no clear consensus for a one size fits all approach [31–35]. Similarly, there are vascular enhancement algorithms to highlight structures that resemble vessels before quantification, but these methods also risk affecting the apparent size of the vessels in the network [34, 36, 37].

Furthermore, OCT is prone to artefacts which specifically affect visualisation of the vasculature [38–40]. When viewed in cross-section, large vessels often demonstrate an hourglass appearance with stronger scattering in the centre than at the sides. This appearance is due to orientation effects of RBCs, which align with the blood flow, causing a reduced scattering cross-section at the sides of the vessels [38]. We hypothesise that these artefacts may lead to underestimation of measurements of vessel diameter. To compensate for these artefacts, OCT contrast agents such as Intralipid have previously been used to increase intravascular intensity signals, particularly in regions where RBC scattering is diminished [39–43]. **Angiography from imaging data predates OCTA. Fluorescein and indocyanine green angiography both image the vasculature after dye injection, using either fundus cameras or scanning laser ophthalmoscopes [44, 45], and X-ray coronary angiography images the lumen after injection of a radio-opaque agent [46]. In each case the contrast is from an agent that fills the vascular lumen rather than from the scattering by the red blood cells, so these methods are not subject to the RBC orientation artefact. However, they lack the depth resolution provided by OCT.**

In this study we examine different methods of measuring vessel diameter from in vivo OCT scans of mouse retinal vasculature, and we use a contrast agent to obtain ground truth vessel diameter measurements for comparison. In addition to traditional binarisation-based methods, we also employ model-based methods for quantifying vessel diameter. Traditional Gaussian models have been employed for vessel diameter measurements in techniques such as fluorescein angiography [47], fundus photography [48], and OCT [49], but we expand this by also investigating the generalised Gaussian model. As part of this study, we measure how much diameter is underestimated from en face projections and evaluate how reliably it can be compensated through calibration. Flicker stimulation is well established to evoke vasodilation in the retina [50–56]. After identifying the best methods for quantifying diameter and compensating for underestimations, we finally evaluated a test scenario using flicker stimulation to induce a measurable change in diameter to evaluate the performance of different methods.

2. Methods and Materials

2.1. Imaging system

All imaging was performed using a custom-built polarisation-sensitive spectral-domain OCT ophthalmoscope, described in detail previously [57]. The spectrum was centred at $\lambda = 840$ nm with a spectral bandwidth of 100 nm at full width half maximum (FWHM), providing an axial resolution of approximately 5.1 μm in air and 3.8 μm in tissue (assuming a refractive index of 1.35). A sensitivity of 96 dB was achieved at an A-scan rate of 80 kHz and 2.85 mW of power measured at the cornea. A beam diameter of $D = 0.5$ mm was used to avoid vignetting effects at the pupil plane [58]. For an aberration-free mouse eye of focal length $f = 2.6$ mm [59,60], this beam diameter gave a theoretical lateral resolution at the retina of $4\lambda f/(\pi D) = 5.6$ μm as the $1/e^2$ diameter [61]. This system was employed to acquire volumetric OCT data from mouse retinas before, during, and after an intravascular contrast agent administration, as well as before and after functional stimulation of the mouse retina using a custom 501 nm light-emitting diode (LED) ring flashing at 10 Hz with a FWHM of 30 nm (power at cornea = 26.6 μW). This wavelength was selected to target the peak absorption of mouse rhodopsin (~ 500 nm), enabling predominantly rod-driven photoreceptor stimulation [62].

2.2. Experimental protocol

Four mice were used in this study. In keeping with the 3Rs of animal research (replacement, reduction and refinement) [63], data from two animals imaged for previously published studies were reused: one very-low-density-lipoprotein receptor (VLDLR) knockout mouse (JAX Stock #002529, B6;129S7-Vldlr^{tm1Her}/J, The Jackson Laboratory, Bar Harbor, USA) [58], and one non-transgenic littermate from a superoxide dismutase 1 (SOD1) knockout mouse (JAX stock #002972, maintained on C57BL/6J) [64]. The remaining two animals were C57BL/6J mice (Stock ID SC-C57J-F, C57BL/6JRj, Janvier Labs, Saint-Berthevin, France) and were imaged separately for this study.

Intralipid 20% (3 mL/kg body weight) was used as a contrast agent and was administered as a bolus injection via the tail vein. A dynamic-contrast OCT (DyC-OCT) protocol [43] was employed during the contrast agent injection and during stimulation, consisting of 500 A-scans and 2000 B-scans repeated over approximately 16 seconds (interscan time 8.2 ms) at the same cross-sectional location over a ~ 1 mm lateral field of view. DyC-OCT scans allow us to track the same vessels over time to precisely measure the effects of the contrast agent or the stimulation with a high degree of spatial coregistration. All quantitative vessel diameter measurements reported in this study were derived from the DyC-OCT data. Additionally, volumetric angiography data were acquired over a ~ 1 mm $\times$ 1 mm field of view centred on the optic nerve head (ONH), with a scan density of 500 A-scans per B-scan and five repeated B-scans at each of 400 slow-axis positions (acquisition time approximately 16 seconds). Pre- and post-injection volumetric datasets were acquired and used for qualitative analysis.

As an exploratory proof-of-concept, functional flicker stimulation was performed in one of the C57BL/6J mice to assess whether the choice of diameter measurement method affects the detection of physiological vascular responses. The mouse was dark-adapted for approximately 2 hours prior to imaging. During the DyC-OCT acquisition, the retina was unstimulated for approximately 7 seconds, followed by 10 Hz flicker stimulation from the LED ring for the remainder of the scan, approximately 9 seconds. To compare results before and after stimulation, the data were divided into baseline (pre-stim) and post-stim. Considering the time required for the vascular functional response to stabilise after stimulation, four-second time windows at the beginning and end of each scan were selected for the pre- and post-stimulation analyses respectively.

All imaging sessions were conducted under general anaesthesia. Anaesthesia and handling for

141 the VLDLR knockout and SOD1 non-transgenic littermate mice are described in our previous
142 publications [64, 65]. Both C57BL/6J mice were initially induced with isoflurane at 4% in
143 oxygen for 4 minutes and were subsequently maintained under different anaesthetic combinations.
144 The mouse used for flicker stimulation received a ketamine + xylazine cocktail (100 mg/kg
145 ketamine, 6 mg/kg xylazine), chosen over continued isoflurane to minimise anaesthesia-induced
146 vasodilation [66, 67], while the other mouse received a combination of medetomidine + midazolam
147 + fentanyl + ketamine (MMFK, 10 mL/kg body weight). To maintain core body temperature
148 throughout imaging, the mice were positioned on a heating pad. The pupils were dilated with
149 one drop of tropicamide (5 mg/mL, Agepha Pharma s.r.o., Senec, Slovakia) per eye, and artificial
150 tear drops (Oculotect, ALCON Pharma GmbH, Freiburg im Breisgau, Germany) were applied at
151 regular intervals to prevent corneal drying.

152 All animal procedures were approved by the animal ethics committee of the Medical Uni-
153 versity of Vienna and the Austrian Federal Ministry of Education, Science, and Research
154 (BMBWF/66.009/0272-V/3b/2019 and GZ 2024-0.802.524).

155 2.3. *Post-processing and analysis*

156 Cross-polarised OCT intensity data from the retinal pigment epithelium (RPE) was used for
157 inter-frame alignment and B-scan flattening [58, 68]. Angiograms were derived by applying
158 the complex subtraction OCT angiography (OCTA) algorithm [22]. Multiple combinations
159 of methods for quantifying vessel diameter from the OCTA data were evaluated to determine
160 which combinations of steps work best under different conditions. Specifically, two types of
161 data projections (maximum intensity and mean intensity) to generate vessel profiles and several
162 different quantifications of those profiles including a traditional global threshold, a Gaussian
163 model approach, and a generalised Gaussian (GG) model approach were evaluated. The projection
164 and global threshold steps followed established OCTA quantification practice [8, 9, 69], whereas
165 the model-based approaches built on Gaussian profile fitting previously applied to retinal vessel
166 diameter measurement in other optical imaging modalities and in OCT [47–49], which we
167 extended to the GG model. The different methods were evaluated against the ground truth as
168 described in this section. All data processing and analysis was performed in MATLAB (R2019b,
169 MathWorks, Natick, MA, USA).

170 2.3.1. Ground truth calculation

171 Ground truth vessel diameters were obtained from contrast-enhanced DyC-OCT B-scan an-
172 giograms. The lateral (d_x) and axial (d_z) diameters of the vessel were manually measured
173 from the vessel's cross-section following the arrival of the Intralipid bolus. The ground truth
174 diameter was defined as the minimum of the two measurements, $d_{GT} = \min(d_x, d_z)$, to account
175 for potential off-axis cross-sectioning of the vessel, which produces a more elliptical appearance
176 in the B-scan. This assumes a circular vascular cross-section for a vessel oriented perpendicular
177 to the imaging plane. For a vessel perpendicular to the B-scan, the lateral and axial diameters are
178 in agreement.

179 2.3.2. Vessel projection calculation

180 For each large vessel in the field of view, a rectangular region of interest (ROI) encompassing
181 the vessel cross-section, along the lateral (x) and axial (z) dimensions, was selected from the
182 DyC-OCT B-scan angiograms. The ROI was deliberately extended to include a margin of
183 surrounding tissue around the vessel, so that the profile contained a baseline on either side of
184 the vessel for the model fits. En face angiogram profiles were then generated by projecting the
185 linear angiogram intensity within this ROI along the axial (z) direction for all time points. Two
186 projection methods were compared: maximum intensity projection (MIP), which selects the
187 highest intensity value along the depth axis of each ROI, and mean intensity projection (MeIP),

188 which averages the values along each ROI. Each projection resulted in a time-stack of 1-D lateral
 189 intensity profiles for each vessel, which were then used for subsequent diameter quantification.

190 2.3.3. Skew correction

191 Off-axis cross-sectioning elongates the vessel along the lateral direction, so the lateral diameter
 192 measured from the ROI projection can overestimate the true vessel size. Because the lateral
 193 diameter is the quantity studied here, a correction factor of d_{GT}/d_x was defined from the ground
 194 truth diameter. We multiplied this correction factor against the diameter measured from each
 195 method to correct the measured vessel diameters for the effect of off-axis cross-sectioning.

196 2.3.4. Model-based vessel diameter calculation

197 For the model-based approaches, the vessel projection profiles were normalised to a range of 0–1
 198 to ensure a good fit. Two parametric models, the standard Gaussian and generalised Gaussian
 199 (GG) models, were fitted (Fig. 1a) to the one-dimensional intensity profiles using nonlinear
 200 least-squares regression in MATLAB.

201 Standard Gaussian function:

$$f(x) = A \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) + C \quad (1)$$

202 where A is the peak amplitude, μ is the centre position, σ is the standard deviation, and C is a
 203 constant baseline offset.

204 Generalised Gaussian (GG) function:

$$f(x) = A \exp\left(-\left(\frac{|x - \mu|}{\alpha}\right)^\beta\right) + C \quad (2)$$

205 where A is the peak amplitude, μ is the centre position, α is the scale parameter, β is the shape
 206 parameter, and C is a constant baseline offset. Because the measured profile is the vessel's
 207 reflectivity profile convolved with the system point spread function, it cannot be sharper than
 208 the point spread function itself, and so profiles sharper than Gaussian profile ($\beta < 2$) were not
 209 considered here. The shape parameter β was hence constrained to $\beta \geq 2$ during fitting. With this
 210 constraint, the GG accommodated to profiles ranging from Gaussian ($\beta = 2$, where $\sigma = \alpha/\sqrt{2}$)
 211 to increasingly flat-topped ($\beta > 2$) (Fig. 1a).

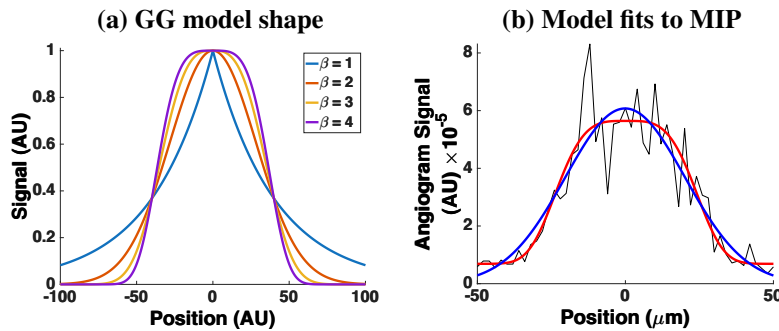


Fig. 1. (a) GG function for shape parameters $\beta = 1, 2, 3$, and 4. (b) Example vessel intensity profile (black) taken from a MIP, with standard Gaussian (blue) and GG (red) fits. The MIP profile is flat-topped ($\beta = 3.7$), a shape the standard Gaussian cannot represent.

Two standard width metrics were calculated from the fitted models: the full width at half maximum (FWHM) and the $1/e^2$ width. For the standard Gaussian fit, the $1/e^2$ width was calculated as:

$$w_{1/e^2, G} = 4\sigma \quad (3)$$

The FWHM for a Gaussian is related to the $1/e^2$ width by a fixed proportionality:

$$w_{fwhm, G} = 2\sigma\sqrt{2\ln 2} = \frac{\sqrt{2\ln 2}}{2} w_{1/e^2, G} \quad (4)$$

and therefore only the $1/e^2$ width was calculated directly from the fit and evaluated here, with the FWHM obtainable via this known relationship.

For the GG fit, no such linear relationship exists between the FWHM and $1/e^2$ width, since both depend on the shape parameter β . Both metrics were therefore calculated independently as:

$$w_{fwhm, GG} = 2\alpha(\ln 2)^{1/\beta} \quad (5)$$

$$w_{1/e^2, GG} = 2\alpha \cdot 2^{1/\beta} \quad (6)$$

2.3.5. Binary-mask-based vessel diameter calculation

For comparison with OCTA processing pipelines that rely on image binarisation, vessel diameters were also measured from binary masks generated by global thresholding. To determine the threshold level, the maximum angiogram intensity was measured after applying a 3D averaging filter with a kernel size of $50 \times 50 \times 10$ voxels (lateral $x \times$ axial $z \times$ temporal t) to the DyC-OCT angiogram volumes to reduce noise and minimise the effects of outliers. To produce a binary mask (BM), the maximum angiogram intensity within these smoothed volumes was calculated, and five different threshold values were defined as fixed percentages of these maxima for MIP and MeIP. For MIP, the thresholds were defined as BM-1 = 40%, BM-2 = 50%, BM-3 = 60%, BM-4 = 70%, and BM-5 = 80%. While for MeIP, the thresholds were defined as BM-1 = 7%, BM-2 = 8%, BM-3 = 9%, BM-4 = 10%, and BM-5 = 11%. These thresholds were selected empirically to cover the usable range of values for each projection. The upper and lower bounds were selected based on the point at which vessels began to be lost, and at which background noise and tissue signal were increasingly included, respectively. Each threshold was applied to the unsmoothed en face projection (MIP and MeIP) to generate a binary mask, and the vessel diameter was measured as the number of pixels above the threshold across the lateral vessel cross-section.

2.3.6. Statistical analysis

To evaluate the accuracy and precision of all diameter measurement approaches, the calculated diameter values were plotted against the ground truth for each vessel across three conditions: pre-contrast, peak contrast, and post-contrast. A linear regression was fitted to each condition, yielding the slope (m) and coefficient of determination (R^2). A slope of $m = 1$ indicated perfect agreement with the ground truth, and R^2 quantified the goodness of fit of the linear model, indicating how well it can be corrected by a calibrated scaling factor. Additionally, the coefficient of variation (CoV) was computed for each vessel across the time samples within each contrast condition and then averaged across vessels, to assess measurement precision (repeatability). For the model-based methods, six combinations were evaluated: two projection methods (MIP and MeIP) by three width metrics ($w_{1/e^2, G}$, $w_{fwhm, GG}$, and $w_{1/e^2, GG}$). For the binary-mask-based method, diameter was measured at five threshold levels (BM-1 through BM-5) for both MIP and MeIP. For the functional flicker stimulation experiment, a paired t-test was used to evaluate whether vessel diameters differed significantly between the pre- and post-stimulation windows,

252 and the Pearson correlation coefficient (ρ) between measured diameter values and angiogram
253 intensity was computed for each method and vessel to assess whether the diameter measurements
254 were affected by changes in angiogram intensity. Measurements with $p < 0.05$ were considered
255 statistically significant (significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

256 3. Results

257 3.1. Qualitative assessment of vessel underestimation

258 Figure 2 shows en face OCTA maximum intensity projections of the superficial vascular plexus
259 (SVP), intermediate capillary plexus (ICP), and deep capillary plexus (DCP) before and after
260 contrast injection. Across all three layers, vessels appeared visibly wider and brighter post-contrast
261 injection, consistent with previous literature [58]. This effect was most pronounced in the larger
262 vessels, especially in the SVP, where the major retinal arteries and veins appeared noticeably
263 thicker after contrast administration. In the ICP and DCP, the capillary networks also showed
264 improved visibility, though the effect was less dramatic.

265 DyC OCTA projections helped visualise the temporal changes in vessel diameter pre- and
266 post-contrast injection (Fig. 3). Pre-contrast injection, the vessel cross-sections appeared narrower
267 than in the contrast-enhanced phase (Fig. 3a), which is due to lateral edge signal suppression
268 by the RBC orientation artefact [40]. At peak contrast, the Intralipid filled the vascular lumen
269 completely. The vessels appeared slightly wider at peak than post-contrast, but this could be due
270 to the dynamic range of the image rather than a true diameter change (Fig. 3b). In the post-contrast
271 phase, the signal stabilised as the initial bolus cleared, and the vessel boundaries remained more
272 visible than before injection (Fig. 3c). The MIP and the corresponding binary mask over time
273 (Figs. 3d,e) showed the observed temporal changes in vessel diameter across the three phases.
274 The mean OCTA intensity (Fig. 3f) showed a sharp increase post-contrast injection, followed by
275 a decrease to a new steady state that remained elevated above the pre-contrast baseline.

276 3.2. Accuracy and precision of diameter measurements

277 All methods underestimated vessel diameter in pre-contrast data, with slopes ranging from 0.42
278 to 0.73 (Table 1). Contrast enhancement improved accuracy across all metrics, though the degree
279 of improvement varied.

280 The $w_{1/e^2, G}$ with MIP slightly overestimated at both peak and post-contrast ($m = 1.15$ and
281 $m = 1.12$), while MeIP yielded values closer to unity ($m = 1.04$ in both conditions). The
282 $w_{1/e^2, GG}$ with MeIP provided the best overall agreement, achieving a slope closest to unity with
283 $m = 1.001$ ($R^2 = 0.97$) post-contrast, which was the closest to the identity line of all tested
284 model-based methods. MeIP generally gave slopes closer to unity than MIP, and also provided
285 smaller standard deviations for the model-based methods (Fig. 4). Pre-contrast R^2 was higher for
286 MIP across all three metrics, and MeIP only surpassed MIP post-contrast (Table 1, Fig. 4).

287 For binary-mask-based methods, the slope decreased with increasing threshold across all
288 contrast conditions and projections (Fig. 5, Fig. 6) (Table 1). MeIP slopes were consistently
289 higher than MIP at every threshold and contrast condition, though at lower thresholds and at peak
290 and post-contrast, MeIP tended to slightly overestimate vessel diameter. The R^2 values showed
291 opposite trends for the two projections, which are most directly compared in the summary bar
292 plots of Fig. 7c,d. For MIP, R^2 decreased with increasing threshold across all contrast conditions,
293 most strongly post-contrast (Fig. 7c). For MeIP, R^2 instead increased with threshold, most evident
294 at peak contrast (Fig. 7d). Overall, MeIP provided more accurate binary-mask-based diameter
295 measurements than MIP at most thresholds, and was more linear at the higher thresholds. The
296 exception was the lowest threshold, where MIP stayed closer to the identity line at peak and
297 post-contrast ($m = 0.96$ and $m = 0.92$, against $m = 1.14$ and $m = 1.11$ for MeIP), and was also
298 more linear (Fig. 5a, Fig. 6a).

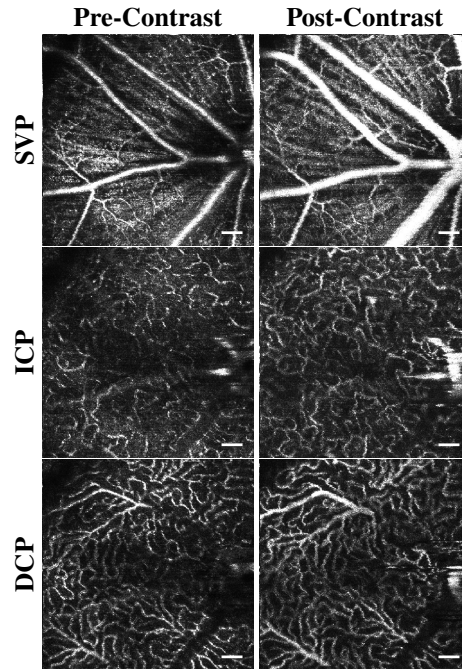


Fig. 2. En face OCTA maximum intensity projections of the SVP, ICP, and DCP before (left) and after (right) the administration of Intralipid 20%. Vessels appear visibly wider and brighter post-contrast across all layers, most prominently in the larger vessels. All scale bars are 100 μm .

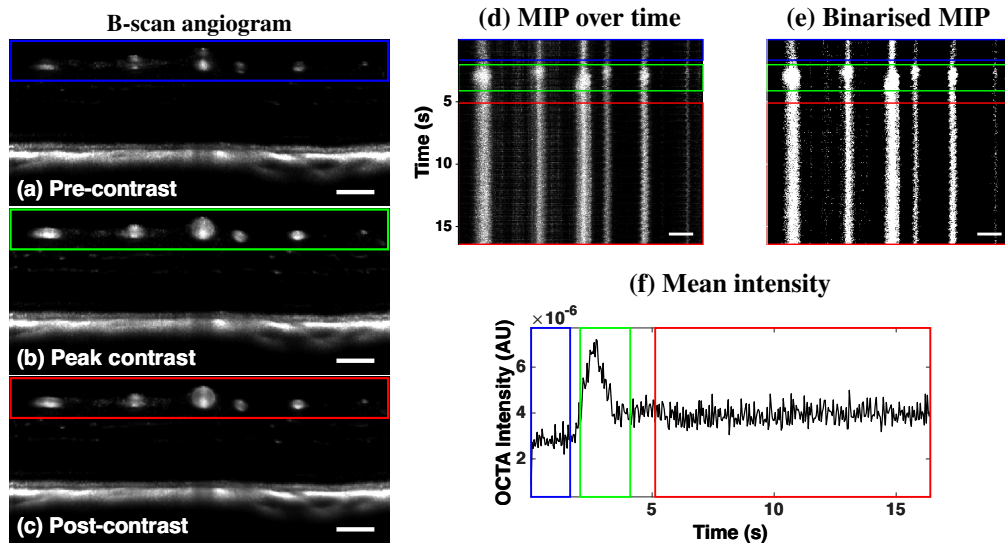


Fig. 3. DyC-OCT B-scan angiograms at (a) pre-contrast, (b) peak contrast, and (c) post-contrast. (d) MIP and (e) binarisation of the vessel cross-sections over time. Coloured boxes indicate pre-contrast (blue), peak-contrast (green), and post-contrast (red) windows. (f) Mean OCTA intensity over time. All scale bars are 100 μm .

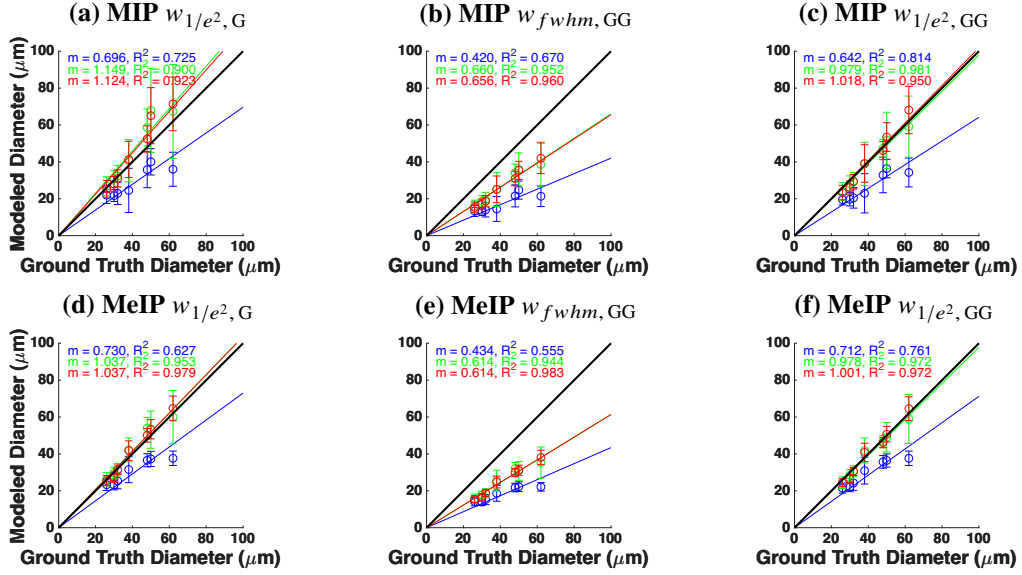


Fig. 4. Modelled versus ground truth vessel diameter using MIP (top row, a–c) and MeIP (bottom row, d–f). Columns correspond to $w_{1/e^2, G}$ (a, d), $w_{fwhm, GG}$ (b, e), and $w_{1/e^2, GG}$ (c, f). Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions. The black line indicates the identity line ($m = 1$).

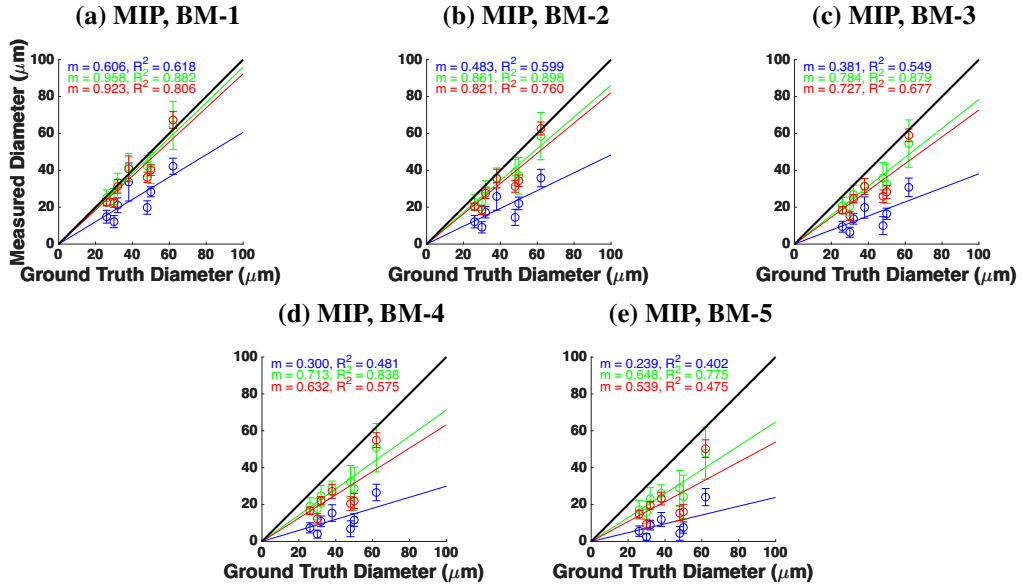


Fig. 5. Vessel diameter measured from binarised masks versus ground truth for MIP at five threshold levels: (a) BM-1 through (e) BM-5. Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions.

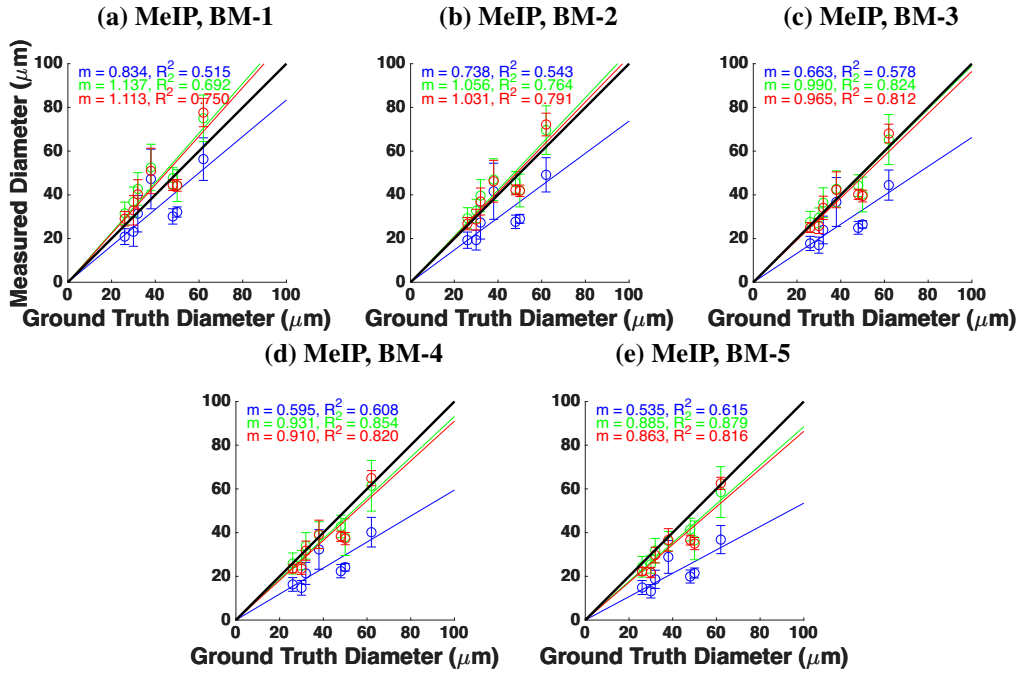


Fig. 6. Vessel diameter measured from binarised masks versus ground truth for MeIP at five threshold levels: (a) BM-1 through (e) BM-5. Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions.

Table 1. Summary of slope (m), coefficient of determination (R^2), and coefficient of variation (CoV) for all diameter measurement methods across pre-contrast, peak contrast, and post-contrast conditions. Methods are grouped as binary-mask-based and model-based. Bold values indicate slope agreement within 10% of unity ($0.9 \leq m \leq 1.1$), strong linearity ($R^2 \geq 0.90$), or high precision (CoV ≤ 0.10).

BM-1			BM-2		BM-3		BM-4		BM-5		$w_{1/e^2, G}$		$w_{fwhm, GG}$		$w_{1/e^2, GG}$	
MIP	MeIP		MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP
<i>Slope (m)</i>																
Pre	0.61	0.83	0.48	0.74	0.38	0.66	0.30	0.60	0.24	0.54	0.70	0.73	0.42	0.43	0.64	0.71
Peak	0.96	1.14	0.86	1.06	0.78	0.99	0.71	0.93	0.65	0.89	1.15	1.04	0.66	0.61	0.98	0.98
Post	0.92	1.11	0.82	1.03	0.73	0.96	0.63	0.91	0.54	0.86	1.12	1.04	0.66	0.61	1.02	1.00
<i>Coefficient of determination (R^2)</i>																
Pre	0.62	0.51	0.60	0.54	0.55	0.58	0.48	0.61	0.40	0.62	0.73	0.63	0.67	0.55	0.81	0.76
Peak	0.88	0.69	0.90	0.76	0.88	0.82	0.84	0.85	0.78	0.88	0.90	0.95	0.95	0.94	0.98	0.97
Post	0.81	0.75	0.76	0.79	0.68	0.81	0.58	0.82	0.48	0.82	0.92	0.98	0.96	0.98	0.95	0.97
<i>Coefficient of variation (CoV)</i>																
Pre	0.20	0.20	0.24	0.19	0.30	0.19	0.37	0.19	0.47	0.19	0.26	0.14	0.27	0.15	0.25	0.13
Peak	0.17	0.16	0.20	0.17	0.22	0.17	0.27	0.17	0.32	0.17	0.24	0.15	0.23	0.16	0.21	0.14
Post	0.10	0.13	0.10	0.12	0.12	0.11	0.15	0.10	0.18	0.09	0.18	0.10	0.18	0.11	0.16	0.09

299 The precision of each method was assessed via the CoV (Table 1, Fig. 7a,b). For MIP, the
 300 binary-mask-based CoV increased substantially with threshold, whereas the model-based CoV
 301 remained stable across all contrast conditions. For MeIP, the binary-mask-based CoV was
 302 notably more stable across thresholds than for MIP, and the model-based CoV values were lower
 303 overall, though they increased slightly at peak contrast before decreasing post-contrast. Across
 304 both methods, MeIP yielded lower CoV values than MIP pre-contrast and more stable values
 305 post-contrast.

306 Among the model-based metrics, $w_{fwhm, GG}$ showed the largest underestimation in all three
 307 contrast conditions, with slopes between 0.42 and 0.66 across both projections. Its linearity was
 308 not correspondingly poor, however, post-contrast it reached $R^2 = 0.96$ for MIP and $R^2 = 0.98$
 309 for MeIP, the highest and joint-highest of the model-based metrics respectively, and its precision
 310 remained comparable to the other model metrics. Overall, model-based methods outperformed
 311 binary-mask-based methods in both accuracy and precision, with two exceptions. At the two
 312 lowest MIP thresholds the binary-mask-based CoV was lower than that of any model-based
 313 metric (0.10 at BM-1 and BM-2 post-contrast, against 0.16–0.18), so MIP was more precise
 314 there but less linear ($R^2 = 0.81$ against 0.92–0.96). At the higher MeIP thresholds precision was
 315 comparable to the model-based methods, though accuracy was lower.

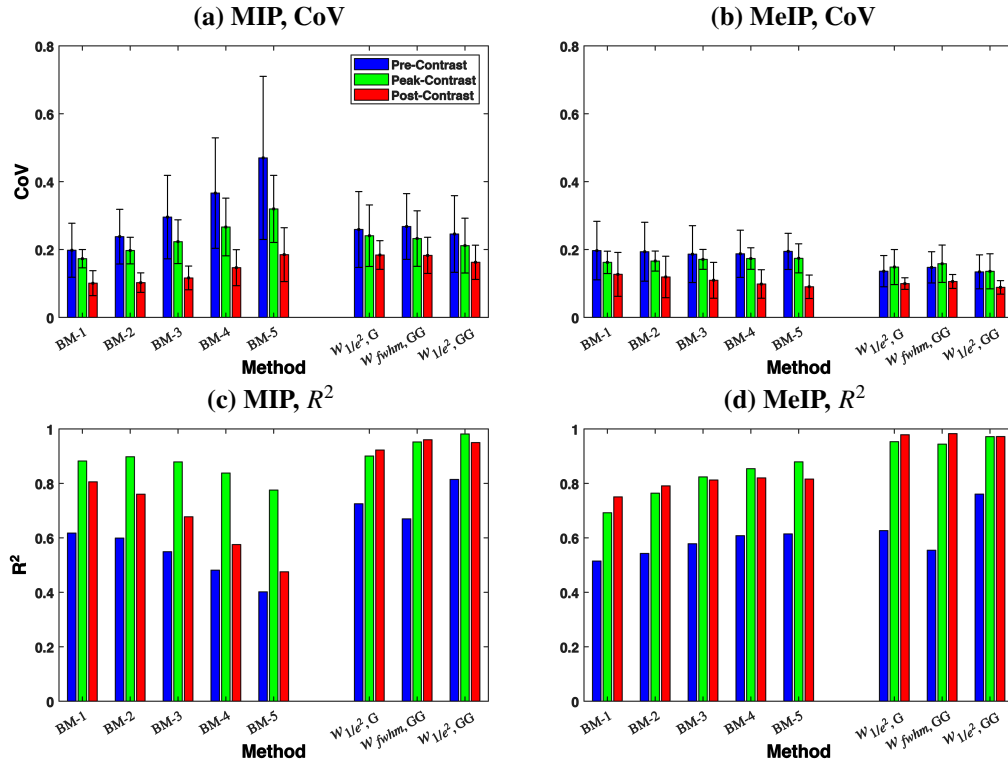


Fig. 7. Summary of linearity and precision across all diameter measurement methods. (a, b) Coefficient of variation (CoV), a measure of precision, and (c, d) coefficient of determination (R^2), a measure of linearity and hence of how reliably a measurement can be corrected, for MIP (a, c) and MeIP (b, d), respectively. Bars are grouped by method (BM-1 through BM-5, $w_{1/e^2, G}$, $w_{fwhm, GG}$, $w_{1/e^2, GG}$) and colour-coded by contrast conditions.

3.3. Functional flicker stimulation

To evaluate how each method performed under a controlled induction of diameter change, flicker stimulation was used to evoke vasodilation while the same vessels were tracked over time. For the binary-mask-based measurements, BM-5 applied to the MeIP was used, as this combination gave the lowest CoV (0.09) and the joint-highest R^2 (0.82) post-contrast. Its slope was further from unity ($m = 0.86$), but that bias is systematic and correctable by calibration. The model-based measurements showed significant diameter increases in four of five vessels ($p < 0.01$), evident in the time series and confirmed by the mean diameters before and after stimulation (Figs. 8a,b,d,e and Table 2). The binary-mask-based time series showed a mixed trend, but the mean diameters revealed significant decreases in four of five vessels (Figs. 8c,f). Of the three methods, the GG model agreed most closely with the ground truth (Figs. 8d–f).

To assess whether angiogram intensity itself biases the diameter measurements, we examined the intensity time series alongside the diameter time series and looked at intensity correlations. The binary-mask-based method showed uniformly positive per-vessel correlations, with an overall correlation of $\rho = 0.526$ (Fig. 9c) measured as the average of the individual correlations. The Gaussian and GG models yielded overall correlations of $\rho = -0.101$ and $\rho = -0.098$, respectively (Fig. 9a,b). The angiogram intensity time series (Fig. 8g) showed a visible decline in angiogram intensity throughout the duration of the DyC-OCT scan across all five vessels. The mean intensity comparison pre- and post-stimulation (Fig. 8h) confirmed that the decrease was significant for all vessels, ranging from -13.6% to -19.8% (Table 2).

Table 2. Flicker stimulation results. Δd : diameter change (%); ρ : Correlation between diameter and angiogram intensity; ΔI : intensity change (%). Statistically significant values are bold. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	Gaussian		GG		Binarised		Intensity	Ground Truth
	Δd (%)	ρ	Δd (%)	ρ	Δd (%)	ρ	ΔI (%)	Δd (%)
Vessel 1 (Artery)	+8.2***	-0.07	+3.5**	0.03	-12.5***	0.59***	-13.6***	+7.9
Vessel 2 (Artery)	+9.9**	0.23*	+10.1***	0.07	-10.2***	0.80***	-18.2***	+21.4
Vessel 3 (Vein)	+1.2	-0.28**	-0.4	-0.28**	-4.1**	0.27**	-14.3***	+0.0
Vessel 4 (Artery)	+28.5***	-0.15	+28.4***	-0.14	+1.0	0.46***	-19.8***	+12.5
Vessel 5 (Vein)	+4.9***	-0.24*	+3.1**	-0.18	-9.3***	0.52***	-16.3***	+7.1
Overall ρ		-0.101		-0.098		0.526		
Overall arterial change	+15.5		+14.0		-7.2		-17.2	+13.9
Overall venous change	+3.1		+1.4		-6.7		-15.3	+3.6

4. Discussion

4.1. Model-based vs. binary-mask-based diameter measurement

The model-based and binary-mask-based diameter methods define vessel boundary in fundamentally different ways. Binarisation of the en face maps discards the spatial intensity distribution for a simple binarisation and reduces each pixel to a binary decision. The measured diameter directly depends on the chosen threshold value and on angiogram intensity, both of which may vary across imaging sessions, contrast conditions, and even between vessels within the same scan. Model-based methods fit a parametric model function to the maximum or mean intensity profiles and calculate the diameter from the fitted parameters. As the fit captures the shape of the profile rather than the absolute intensity, it is inherently less sensitive to changes in angiogram signal, and the level at which the diameter is defined is intensity-independent rather than set by the absolute intensity of the image.

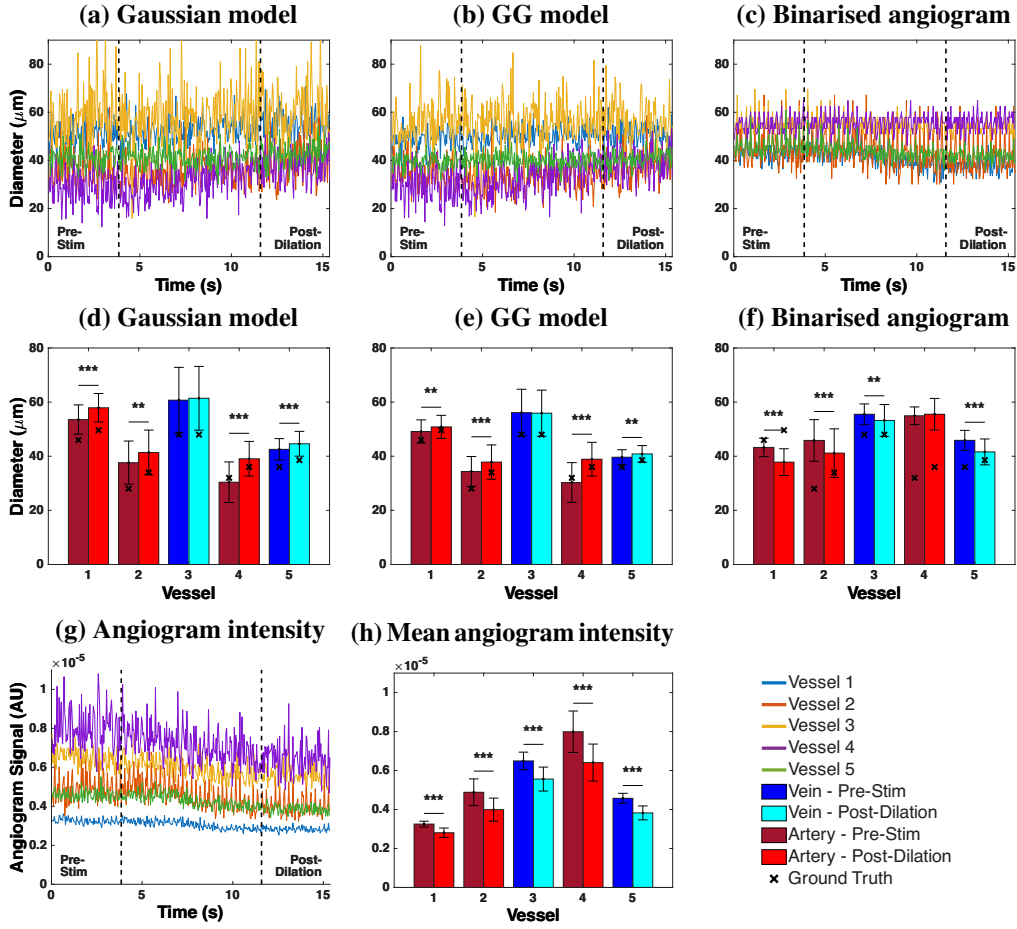


Fig. 8. Functional flicker stimulation results. (a–c) Vessel diameter time series pre- and post-stimulation across five vessels. (d–f) Mean diameter per vessel before (dark bars) and after (light bars) stimulation for each method. Veins are shown in blue/cyan and arteries in dark/light red. (g) Angiogram intensity time series. (h) Mean angiogram intensity per vessel pre- and post-stimulation. All measurements used MeIP. Crosses (x) indicate the ground truth diameter for each vessel. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

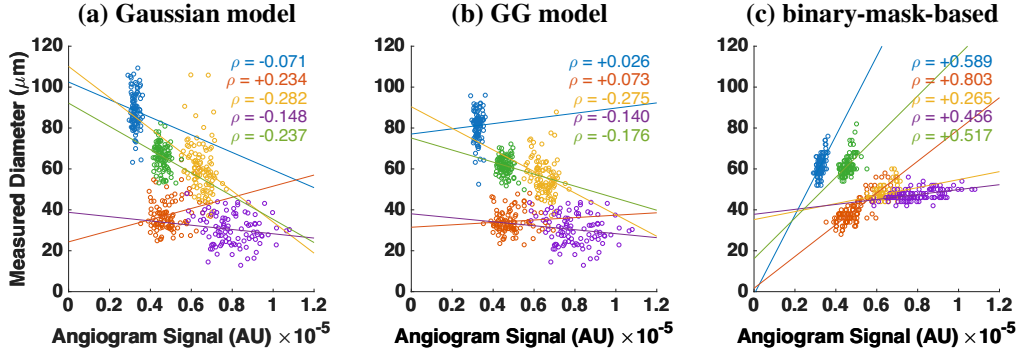


Fig. 9. Measured vessel diameter as a function of angiogram signal intensity during functional flicker stimulation for (a) Gaussian model, (b) generalised Gaussian model, and (c) binary-mask-based method. Each colour represents one of five vessels. ρ denotes the Pearson correlation coefficient.

A practical consequence of this difference is how correctable each method is. Although both methods take input parameters to determine the measured diameter, the model-based diameter is determined relative to the fitted profile and is therefore intensity-independent. In contrast, the binarisation slope shifts with vessel intensity, so any calibration must also account for the intensity threshold used. Systematic diameter underestimation can be corrected by a fixed proportion using the slope, provided the relationship with the ground truth is strongly linear, but such a rescaling removes only the systematic proportional bias and leaves the scatter about the regression unchanged. Linearity, rather than slope alone, therefore determines how reliably a method can be corrected. Post-contrast every model-based metric reached $R^2 \geq 0.92$, whereas binary-mask-based MIP, which is the most commonly used [8,9], ranged from 0.48 at BM-5 to 0.81 at BM-1. The improved linearity of the model-based metrics makes them more predictable/systematic even pre-contrast, where $w_{1/e^2, GG}$ retained $R^2 = 0.81$ against 0.62 for the best binary-mask-based method.

This difference explains the trends observed in Fig. 7 and Table 1. Post-contrast, every model-based metric was more linear than every binary-mask-based method, and the GG $1/e^2$ metric reached near-unity slopes at peak and post-contrast without an intensity threshold parameter to tune, whereas for the binary-mask-based methods linearity was overall lower and depended on threshold using the global thresholding described here. Because the peak angiogram intensity varies between vessels, a fixed threshold removes a different fraction of the cross-section from each one. The measured diameters therefore vary from the ground truth by different amounts, which weakens the linearity. Pre-contrast, the hourglass artefact caused by RBC orientation within vessel lumen suppresses edge signals [38,40], causing all methods to underestimate diameter. However, model-based methods still retained more of that linearity, producing systematic, predictable underestimation. Binary-mask-based methods, on the other hand, showed weaker linearity in most conditions, potentially because the suppressed edge signals did not consistently exceed the binarisation threshold, and because different vessels have different intensity.

The same mechanism accounts for how each method responded to the change in angiogram intensity from peak to post-contrast. Angiogram intensity is highest as the bolus arrives and then settles to a lower steady state (Fig. 3f), so a fixed threshold encloses a smaller fraction of each cross-section post-contrast than at peak, an effect visible in the binarised cross-sections over time (Fig. 3d,e). Model-based metrics were largely insensitive to this, changing by at most 0.04 in slope and 0.04 in R^2 between the two conditions. Binary-mask-based measurements from MIP degraded progressively with threshold, the slope falling by 0.04 at BM-1 but 0.11 at BM-5,

381 and R^2 by 0.07 and 0.30 respectively, while MeIP remained comparatively stable (slope change
382 0.02–0.03 at every threshold). Model-based methods therefore showed comparable agreement
383 with the ground truth at both peak and post-contrast, whereas the agreement of binary-mask-based
384 method depended on the combination of threshold and contrast condition.

385 In this study we observed that the generalised Gaussian (GG) model outperformed the standard
386 Gaussian, because the intensity profiles of the vessels were usually not purely Gaussian (Fig. 1b).
387 The standard Gaussian cannot capture relatively flat-topped angiogram intensity profiles and
388 tends to fit a narrower curve, whereas the GG model accommodates through the shape parameter
389 β (Fig. 1b). The flat-topped shapes arise because the measured profile is effectively the vessel's
390 reflectivity profile convolved with the system point spread function. Because the GG reduces to a
391 standard Gaussian when $\beta = 2$ and departs from it only when the data support a flatter profile, a
392 correctly applied GG fit should perform at least as well as the Gaussian, at the cost of additional
393 computation.

394 Among the width metrics, $w_{fwhm,GG}$ consistently showed the greatest underestimation,
395 although its linearity remained high post-contrast ($R^2 = 0.96$ for MIP and 0.98 for MeIP),
396 indicating that the underestimation is systematic and therefore correctable. Near the half-
397 maximum the profile shape changes rapidly and varies between vessels, from Gaussian to
398 flat-topped, so the width measured at that level is more variable. The $1/e^2$ width captures more
399 of the transition region and so may have provided a more accurate measurement of the vessel
400 diameter. These results suggest that the generalised Gaussian $1/e^2$ metric was the most reliable
401 of the model-based options, showing both the closest agreement with the ground truth, strongest
402 linearity, and highest precision.

403 4.2. Maximum vs. mean intensity projection

404 Across nearly all methods and conditions, MeIP outperformed MIP in both accuracy and precision
405 (Table 1). Linearity was the exception. Pre-contrast MIP produced higher R^2 than MeIP for all
406 model-based metrics, and MIP also produced higher R^2 at the two lowest binarisation thresholds,
407 with MeIP outperforming MIP only post-contrast. This finding is noteworthy because MIP
408 has been shown to produce higher signal-to-noise ratio (SNR) and contrast in en face OCTA
409 images [69], and is the default in many commercial OCTA systems and analysis pipelines [8, 9].
410 However, we observed that higher SNR did not necessarily translate to more accurate diameter
411 measurements. The MIP selects the single highest intensity value along each A-scan within the
412 vessel ROI, which makes it inherently sensitive to outliers. MeIP averages all intensity values
413 along each A-scan, which smooths out intensity fluctuations and provides a consistent profile.
414 This averaging could be beneficial for vessel diameter measurement, as the goal is to capture the
415 typical vessel boundary rather than extreme signal events. This trade-off may also explain why
416 linearity behaved differently before and after contrast. Pre-contrast the vessel signal is weakest,
417 and edge suppression by the RBC orientation artefact leaves little margin above the background,
418 so the higher contrast of the MIP helps keep the profile distinguishable across vessels of different
419 size. Once the lumen is filled with contrast agent, signal is no longer limiting and the outlier
420 sensitivity of a maximum statistic may contribute more strongly to the measurement error, so the
421 averaging of the MeIP yields the tighter relationship with the ground truth. Consistent with this,
422 MeIP produced higher R^2 than MIP for all three model-based metrics post-contrast, whereas at
423 peak contrast the two projections remained comparable, with MIP higher for two of the three
424 model-based metrics.

425 The difference between MIP and MeIP was especially apparent in the binary-mask-based
426 results (Table 1). For MeIP, the slope declined more gradually and remained at 0.86 even at
427 BM-5 post-contrast. Whereas for MIP, the slope decreased steeply with increasing threshold,
428 dropping from 0.92 at BM-1 to 0.54 at BM-5 post-contrast. These findings suggest that MeIP
429 produces a more uniform intensity distribution across the vessel profile, making the binarisation

430 result less sensitive to the exact threshold value. The CoV data (Table 1) reinforced this as the
 431 MIP-based binary mask CoV increased substantially with threshold, whereas the MeIP-based
 432 CoV remained relatively stable. This behaviour follows from how each projection responds to
 433 variance in the angiogram signal. Because the MIP keeps only the brightest pixel along each
 434 A-scan, it tracks whichever value happens to be largest, so the projected profile fluctuates from
 435 one time sample to the next. A threshold applied to a fluctuating profile returns a fluctuating
 436 width. How much the width fluctuates depends on where the threshold cuts the profile. A low
 437 threshold cuts the steep flanks of the profile, where a change in intensity moves the measured
 438 edge only slightly. A high threshold cuts near the flat top, where the same change moves it
 439 much further. The CoV therefore grows as the threshold is raised. MeIP averages along each
 440 A-scan instead, which suppresses the fluctuation and gives a broader, smoother profile, so its CoV
 441 changes little as the threshold increases. For model-based methods, MeIP also showed better
 442 performance. The $w_{1/e^2, G}$ using MIP slightly overestimated vessel diameter post-contrast with
 443 $m = 1.12$, while MeIP reduced this to $m = 1.04$, a much closer agreement with the ground truth.
 444 These results suggest that while MIP may be preferred for visualisation purposes where contrast
 445 and SNR are important [69], MeIP should be the preferred projection method for quantitative
 446 diameter measurements using the methods evaluated here.

447 4.3. Influence of angiogram intensity on diameter measurements

448 The functional flicker stimulation experiment provided a direct demonstration of how changes in
 449 angiogram signal intensity can affect diameter measurements based on the method used. During
 450 the DyC-OCT acquisition, the angiogram intensity declined significantly across all five vessels
 451 (Fig. 8g,h), with decreases ranging from -13.6% to -19.8% (Table 2). The cause of this decline
 452 was not established with certainty, although inspection of the volumetric data ruled out the most
 453 important confounds. There was no obvious out-of-plane motion or breathing artefact, and the
 454 image quality was consistent throughout the scan. We suspect a combination of rapid corneal
 455 drying and/or cataract formation over the course of the scan could have caused the decline in
 456 intensity. More importantly, the cause of the decline does not affect our observation that the
 457 intensity decline was associated with changes in the binary-mask-based measurements, while the
 458 model-based measurements were less affected.

459 The model-based methods correctly detected the expected vasodilation, with the Gaussian
 460 model showing significant diameter increases in four of five vessels ($+15.5\%$ overall in the
 461 arteries against $+3.1\%$ in the veins) and the GG model showing significant increases in four of
 462 five vessels ($+14.0\%$ and $+1.4\%$ respectively; Table 2). The binary-mask-based method, however,
 463 reported significant diameter decreases in four of five vessels, suggesting vasoconstriction which
 464 is a paradoxical result. The reason behind this behaviour could be understood by observing
 465 how each method responds to declining signal intensity (Fig. 9). For the binary-mask-based
 466 method, the threshold is fixed and diameter is determined by the number of pixels that exceed
 467 this threshold. When the overall angiogram intensity drops, fewer pixels at the vessel edges
 468 exceed the threshold, and the vessel appears to shrink even if the physical vessel diameter has
 469 increased. This is confirmed by the strong positive correlation between measured diameter and
 470 angiogram intensity for the binary-mask-based method ($\rho = 0.526$, $p < 0.0005$). In contrast, the
 471 model-based methods showed weak overall correlations with intensity ($\rho = -0.101$ for Gaussian,
 472 $\rho = -0.098$ for GG), indicating that the fitted diameter was effectively independent of the signal
 473 intensity. Among the methods, the GG model also produced the lowest standard deviations across
 474 the five vessels in the flicker experiment (Fig. 8d-f), consistent with its low CoV post-contrast
 475 (Table 1), indicating that it is the most repeatable of the metrics tested. It also showed the closest
 476 agreement with ground truth. Together these results demonstrated that the model-based diameter
 477 calculation provided more accurate and repeatable measurements, and recovered physiologically
 478 meaningful vascular responses that binary-mask-based methods misidentified.

479 These findings have important implications for longitudinal studies and any experiment where
 480 angiogram intensity may vary between or within imaging sessions. In longitudinal mouse eye
 481 imaging, the angiogram signal can vary between and within sessions as anaesthesia alters retinal
 482 perfusion and ocular physiology [67, 70], with additional factors such as cataract formation, and
 483 corneal drying if tear drops are not applied, further reducing OCT intensity over time [71, 72]. If
 484 a binary-mask-based method is used under such conditions, observed changes in vessel diameter
 485 may reflect intensity fluctuations rather than true vascular changes. In contrast, model-based
 486 methods may be less sensitive to these intensity variations, as they characterise the shape of
 487 the intensity profile rather than relying primarily on absolute intensity values to define vessel
 488 boundaries.

489 4.4. Threshold selection

490 The results presented here were based on a basic, global threshold for the binarisation of the
 491 angiogram data. There are numerous other approaches for binarising the angiogram, which will
 492 most likely each behave differently [31–35]. For example, angiogram signals can be enhanced
 493 with vesselness filters [73] and dynamic thresholds can be used to binarise images with varying
 494 SNR. When applying a vesselness filter, it is possible to enhance the shape of the angiogram and
 495 omit noise, but it is also possible for such enhancement to overcompensate the signal, leading
 496 to larger than expected vessel appearance. Measurements obtained with different enhancement
 497 filters have been shown to differ substantially and are not interchangeable [37]. Dynamic
 498 thresholding can also yield a more uniform angiogram appearance, but similarly risks over-
 499 or under-compensating vessels. The threshold chosen is known to alter OCTA quantitative
 500 metrics systematically, with automatic thresholds tending to underestimate relative to a fixed
 501 threshold [31, 35]. Further analysis of these other methods would be required to determine their
 502 efficacy in returning accurate and precise diameter values.

503 The key observation from the present work is that any thresholding approach, regardless of how
 504 the threshold is determined, shares the same fundamental limitation that the measured diameter
 505 depends on the absolute signal intensity at the vessel edges. As demonstrated by the flicker
 506 stimulation experiment, this intensity dependence can introduce artefactual diameter changes
 507 when the angiogram signal varies over time or between sessions. The model-based approach
 508 avoids this limitation by deriving diameter from the shape of the intensity profile rather than
 509 from an intensity cut-off, making it a fundamentally different and more robust alternative for
 510 quantitative diameter analysis.

511 4.5. Best choices under different conditions

512 Under the conditions where a contrast agent is available and accuracy is the primary goal, the
 513 $w_{1/e^2, GG}$ metric using MeIP provides the best results post-contrast. This combination achieved a
 514 slope of $m = 1.001$ with $R^2 = 0.97$ and a CoV of 0.09, yielding the best accuracy, linearity, and
 515 precision of all tested methods. If only pre-contrast data is available, the $w_{1/e^2, GG}$ metric still
 516 provides the best linearity among all tested methods ($R^2 = 0.81$ for MIP, $R^2 = 0.76$ for MeIP),
 517 though the MIP and MeIP slope of $m = 0.64$ and $m = 0.71$ indicates that diameter could be
 518 underestimated by approximately 35% and 30% respectively. In this case, a correction factor
 519 could be applied if absolute diameter values are needed, though relative comparisons between
 520 vessels or timepoints may remain valid without correction.

521 When only MIP data is available, which is the case for many commercial OCTA machines
 522 and existing datasets [8, 9], the $w_{1/e^2, GG}$ still provides good post-contrast accuracy ($m = 1.02$,
 523 $R^2 = 0.95$). For binary-mask-based analysis of MIP data, only the lowest threshold (BM-1)
 524 post-contrast achieved a slope near unity ($m = 0.92$), and the slope degraded rapidly with
 525 increasing threshold values. This highlights that if binarisation must be used with MIP data, a
 526 lower threshold value is essential, though this may come at the cost of increased noise sensitivity.

527 The improved linearity and repeatability of the model-based approaches come at the cost
528 of computation time, as fitting a model is considerably more expensive than binarisation. In
529 our testing, fitting the generalised Gaussian was approximately 40% slower than the standard
530 Gaussian. This difference is hardware dependent, and because the fits at each cross-section are
531 independent, the computation could be parallelised and accelerated with appropriate hardware.
532 If computation time is a limiting factor and model fitting is not practical, MeIP with a moderate
533 threshold (e.g. BM-3) provides a reasonable alternative. At post-contrast, BM-3 using MeIP
534 achieved $m = 0.96$ and $R^2 = 0.81$, which could be acceptable for many applications. MeIP-based
535 binary-mask-based analysis showed more stable CoV values across all thresholds compared to
536 MIP, making the results less sensitive to the exact threshold choice. This stability makes MeIP
537 the preferred projection for binary-mask-based OCTA.

538 4.6. Literature comparison

539 Hormel et al. [69] showed that MIP produces en face OCTA images with higher SNR and contrast
540 than MeIP, and concluded that MIP should be preferred for OCTA visualisation. Our results
541 reveal a different picture for quantitative vessel diameter measurement. We found that MIP
542 provided better linearity before contrast injection, whereas MeIP performed better after contrast
543 injection (Table 1), although the pre-contrast advantage of MIP came at the cost of repeatability.
544 For the generalised Gaussian $1/e^2$ metric, MeIP gave both a slope closer to unity ($m = 0.71$
545 against 0.64) and roughly half the coefficient of variation (0.13 against 0.25) before contrast, so
546 MeIP is arguably preferable even in that condition. These findings suggest that the projection
547 method best suited for visualisation is not necessarily the best suited for quantification, and that
548 the choice of projection should depend on the intended application.

549 The challenge of standardising OCTA image analysis has been addressed by Sampson et al. [8]
550 and Untracht et al. [9], both of whom emphasised the need for transparent, reproducible analysis
551 pipelines. Our findings add to this discussion by showing that the choice of diameter measurement
552 method introduces a source of variability that is at least as significant as the choice of binarisation
553 threshold. The model-based approach presented here offers a path towards more standardised
554 vessel diameter quantification, as it reduces the dependence on selected thresholds and angiogram
555 intensity, and provides measurements that are more robust across imaging conditions.

556 The RBC orientation artefact and its effect on OCTA vessel appearance have been described
557 by Cimalla et al. [38] and Bernucci et al. [40], who used Intralipid contrast to demonstrate that
558 the hourglass pattern in vessel cross-sections is caused by the directional scattering properties of
559 RBCs rather than by true variations in flow. Our work extends this observation by quantifying
560 the degree to which this artefact affects diameter measurements and by proposing model-based
561 fitting as a practical correction strategy. While contrast enhancement with Intralipid is effective
562 for revealing the true vessel diameter, it requires a contrast injection and is not yet applicable in
563 clinical settings. Using a calibrated model-based approach provides a way to partially recover
564 the underestimated diameter from native OCTA data without the need for a contrast agent.

565 In our study, the generalised Gaussian model detected a mean vasodilation of +14% in the
566 arteries and +1.4% in the veins (Table 2). The arteries (Vessels 1, 2, and 4) showed increases of
567 +3.5%, +10.1%, and +28.4%, while the veins (Vessels 3 and 5) showed a smaller response, with
568 +3.1% in Vessel 5 and no significant change in Vessel 3. The standard Gaussian gave a broadly
569 similar picture, with arterial increases of +8.2%, +9.9%, and +28.5% and a venous increase of
570 +4.9% in Vessel 5. These results are comparable to other OCT based rodent studies such as Son
571 et al. [74] and Radhakrishnan and Srinivasan [75]. Of these, Son et al. used the most similar
572 setup to ours, with the same mouse strain (C57BL/6J), the same anaesthetic (ketamine/xylazine),
573 and a comparable flicker stimulus (10 Hz, 504 nm), though a different stimulation period of
574 5 s [74]. Using functional OCT, they reported that the response is anisotropic, with larger percent
575 changes in the axial direction than in the lateral direction. Because the diameters reported here

are lateral, their lateral values offer the closest comparison: +10.8% in the arteries and +8.9% in the veins, against +14.0% and +1.4% respectively for the GG model in our study. The arterial responses are therefore of comparable magnitude, whereas our venous response is substantially smaller. Whether this asymmetry reflects a genuine feature of the murine flicker response, or the small number of vessels sampled remains to be established. The authors did not describe how vessel diameters were quantified from their B-scans, so it is worth considering whether their lateral measurements may have been affected by the intensity-dependent artefacts characterised here. The study by Radhakrishnan and Srinivasan [75] measured arterial dilation in the rat inner retina during a 10 s, 12 Hz light stimulus, reporting a 95% confidence interval of +3.3% to +11.0% ($p < 0.005$) for diameters quantified manually along the arterial tree. Polak et al. [76] showed in humans that the diameter response is relatively stable across flicker frequencies from 4 to 40 Hz, suggesting that frequency differences between studies are unlikely to explain the variation. Vascular response maybe impacted by species, anaesthesia, stimulation frequency and wavelength, stimulus power and delivery method, stimulation duration, dark adaptation period, and imaging modality.

4.7. *Future directions*

A future direction of this work is to move towards fully automated segmentation and calculation of vessel diameters from en face angiogram maps. While ROIs were manually selected in this work, it is also possible to automatically segment angiographic data using traditional skeletonisation methods, which are commonly used for automated OCTA analysis [9, 77, 78]. From a skeletonised image, vessel angle can be computed for further extraction of vascular cross-sections perpendicular to the vessel axis. These cross-sections could then be processed using the model-based methods described here to enable fully automated analysis of 3D OCTA data sets, rather than the semi-automated approach shown here for 2D OCTA data. This type of model-based approach is particularly important for comparing 3D volumes acquired at different time points in the mouse eye, as there are often intensity losses caused by drying of the eye, cataract formation, etc. As we have shown in this work, even low levels of intensity loss occurring over a short time window can significantly affect the measured vessel diameter when using a binary-mask-based approach, but not when using the model based approach presented here.

Additionally, this work focused on the larger vessels of the superficial vascular plexus. Extending the model-based approach to the intermediate and deep capillary plexuses, where vessels are smaller and signal levels are lower, would be valuable for studying microvascular changes in diseases. For all vessels, the measured OCT intensity profile is a convolution of the true vessel profile with the point spread function (PSF) of the incident beam on the retina. Since the lateral resolution of OCT is typically larger than capillary diameters because it is determined by the beam optics rather than the source bandwidth, this convolution becomes increasingly significant for smaller vessels. If the PSF were known, a deconvolution approach could be used to recover the true vessel profile before model fitting [79], potentially improving accuracy for capillary-scale vessels. However, characterising the retinal PSF in vivo remains challenging, as it depends on the ocular optics of each individual eye and may vary with retinal depth and eccentricity.

A physical model built from an idealised anatomy and physiology could in theory be used to predict the vessel diameter more accurately than calibration. However, such a model would be significantly more complex than the models presented here. A vessel of circular cross-section carrying a parabolic flow profile could be taken as the starting geometry. The OCTA signal that the system would measure across the vessel could then be calculated, projected, and quantified with the same methods used here, so that the diameter returned by each method would follow from the imaging process itself. This model would have to incorporate the lateral resolution of the system, the RBC orientation artefact [38, 40], the relationship between the scan speed and the

625 blood flow speed on the resulting OCTA signal [80], and OCT fringe washout, which varies as a
626 function of RBC velocity [81, 82]. Further work would be required to develop and validate such
627 a model.

628 5. Conclusion

629 This study characterised different techniques for quantifying vessel diameter in OCTA and
630 evaluated how underestimation resulting from OCTA artefacts can be corrected. We established
631 ground truth from manual diameter measurement in the in vivo mouse retina after Intralipid
632 contrast enhancement. Binary-mask-based diameter calculation was then compared against
633 model-based profile fitting on both maximum and mean intensity projections. Every method
634 underestimated vessel diameter before contrast. Under pre-contrast conditions, the generalised
635 Gaussian $1/e^2$ method retained the strongest linear relationship with the ground truth, making
636 the residual underestimation correctable by calibration. Post-contrast, the same method applied
637 to the mean intensity projection provided results closest to the ground truth of all combinations
638 tested, with the highest precision of the model-based methods ($CoV = 0.09$). Binary-mask-based
639 measurements were highly intensity-dependent, with a higher degree of correlation compared to
640 the model-based approaches. The practical consequence of that dependence was demonstrated by
641 the flicker stimulation results. As the angiogram signal declined over the course of the acquisition,
642 binary-mask-based diameters followed it and reported vasoconstriction. The model-based fits
643 were effectively independent of intensity and recovered the expected vasodilation. We therefore
644 recommend a generalised Gaussian fit to the mean intensity projection for quantitative OCTA
645 vessel diameter measurement, with a calibrated correction applied where contrast enhancement
646 is not available.

647 6. Back matter

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661 References

- 662 1. H. L. Takusagawa, L. Liu, K. N. Ma, *et al.*, "Projection-resolved optical coherence tomography angiography of
663 macular retinal circulation in glaucoma," *Ophthalmology* **124**, 1589–1599 (2017).
- 664 2. T. S. Hwang, M. Zhang, K. Bhavsar, *et al.*, "Visualization of 3 distinct retinal plexuses by projection-resolved optical
665 coherence tomography angiography in diabetic retinopathy," *JAMA Ophthalmol.* **134**, 1411–1419 (2016).
- 666 3. F. Coscas, A. Glacet-Bernard, A. Miere, *et al.*, "Optical coherence tomography angiography in retinal vein occlusion:
667 Evaluation of superficial and deep capillary plexa," *Am. J. Ophthalmol.* **161**, 160–171 (2016).
- 668 4. M. K. Durbin, L. An, N. D. Shemonski, *et al.*, "Quantification of retinal microvascular density in optical coherence
669 tomographic angiography images in diabetic retinopathy," *JAMA Ophthalmol.* **135**, 370–376 (2017).
- 670 5. A. Yarmohammadi, L. M. Zangwill, A. Diniz-Filho, *et al.*, "Peripapillary and macular vessel density in patients with
671 glaucoma and single-hemifield visual field defect," *Ophthalmology* **124**, 709–719 (2017).

- 672 6. S. Tzaridis, M. W. M. Wintergerst, C. Mai, *et al.*, "Quantification of retinal and choriocapillaris perfusion in different
673 stages of macular telangiectasia type 2," *Investig. Ophthalmol. & Vis. Sci.* **60**, 3556–3562 (2019).
- 674 7. W. R. Kwapong, H. Ye, C. Peng, *et al.*, "Retinal microvascular impairment in the early stages of Parkinson's disease,"
675 *Investig. Ophthalmol. & Vis. Sci.* **59**, 4115–4122 (2018).
- 676 8. D. M. Sampson, A. M. Dubis, F. K. Chen, *et al.*, "Towards standardizing retinal optical coherence tomography
677 angiography: a review," *Light Sci. & Appl.* **11**, 63 (2022).
- 678 9. G. R. Untracht, M. S. Durkee, M. Zhao, *et al.*, "Towards standardising retinal OCT angiography image analysis with
679 open-source toolbox OCTAVA," *Sci. Rep.* **14**, 5979 (2024).
- 680 10. A. Y. Kim, Z. Chu, A. Shahidzadeh, *et al.*, "Quantifying microvascular density and morphology in diabetic
681 retinopathy using spectral-domain optical coherence tomography angiography," *Investig. Ophthalmol. & Vis. Sci.* **57**,
682 OCT362–OCT370 (2016).
- 683 11. Y. Ueno, T. Iwase, K. Goto, *et al.*, "Association of changes of retinal vessels diameter with ocular blood flow in eyes
684 with diabetic retinopathy," *Sci. Rep.* **11**, 4653 (2021).
- 685 12. M. K. Ikram, J. C. M. Witterman, J. R. Vingerling, *et al.*, "Retinal vessel diameters and risk of hypertension: The
686 Rotterdam study," *Hypertension* **47**, 189–194 (2006).
- 687 13. A. Tirsi, N. Leung, R. Gupta, *et al.*, "Retinal vessel diameter reductions are associated with retinal ganglion cell
688 dysfunction, thinning of the ganglion cell and inner plexiform layers, and decreased visual field global indices in
689 glaucoma suspects," *Diagnostics* **15**, 1700 (2025).
- 690 14. C. Liang, C. Gu, and N. Wang, "Retinal vascular caliber in coronary heart disease and its risk factors," *Ophthalmic*
691 *Res.* **66**, 151–163 (2022).
- 692 15. J. P. Mynard, A. Kondiboyina, R. Kowalski, *et al.*, "Measurement, analysis and interpretation of pressure/flow waves
693 in blood vessels," *Front. Physiol.* **11**, 1085 (2020).
- 694 16. A. C. Gonçalves Seabra, A. F. Silva, T. Stieglitz, and A. B. Amado-Rey, "In silico blood pressure models comparison,"
695 *IEEE Sens. J.* **22**, 23486–23493 (2022).
- 696 17. R. J. Hernandez, S. Madhusudhan, Y. Zheng, and W. K. El-Bouri, "Linking vascular structure and function:
697 Image-based virtual populations of the retina," *Investig. Ophthalmol. & Vis. Sci.* **65**, 40 (2024).
- 698 18. R. J. Hernandez and W. K. El-Bouri, "Microvascular retinal digital twins from non-invasive clinical images," in
699 *Digital Twin for Healthcare (DT4H 2025)*, vol. 16193 of *Lecture Notes in Computer Science* (Springer, Cham, 2026),
700 pp. 12–22.
- 701 19. D. Huang, E. A. Swanson, C. P. Lin, *et al.*, "Optical coherence tomography," *Science* **254**, 1178–1181 (1991).
- 702 20. A. F. Fercher, C. K. Hitzenberger, W. Drexler, *et al.*, "In vivo optical coherence tomography," *Am. J. Ophthalmol.*
703 **116**, 113–114 (1993).
- 704 21. S. Makita, Y. Hong, M. Yamanari, *et al.*, "Optical coherence angiography," *Opt. Express* **14**, 7821–7840 (2006).
- 705 22. V. J. Srinivasan, J. Y. Jiang, M. A. Yaseen, *et al.*, "Rapid volumetric angiography of cortical microvasculature with
706 optical coherence tomography," *Opt. Lett.* **35**, 43–45 (2010).
- 707 23. Y. K. Tao, A. M. Davis, and J. A. Izatt, "Single-pass volumetric bidirectional blood flow imaging spectral domain
708 optical coherence tomography using a modified Hilbert transform," *Opt. Express* **16**, 12350–12361 (2008).
- 709 24. R. K. Wang, S. L. Jacques, Z. Ma, *et al.*, "Three dimensional optical angiography," *Opt. Express* **15**, 4083–4097
710 (2007).
- 711 25. A. Javed, A. Khanna, E. Palmer, *et al.*, "Optical coherence tomography angiography: A review of the current
712 literature," *J. Int. Med. Res.* **51**, 03000605231187933 (2023).
- 713 26. B. Tan, R. Sim, J. Chua, *et al.*, "Approaches to quantify optical coherence tomography angiography metrics," *Ann.*
714 *Transl. Med.* **8**, 1205 (2020).
- 715 27. P. Zabel, J. J. Kaluzny, M. Wilkosc-Debczynska, *et al.*, "Comparison of retinal microvasculature in patients with
716 Alzheimer's disease and primary open-angle glaucoma by optical coherence tomography angiography," *Investig.*
717 *Ophthalmol. & Vis. Sci.* **60**, 3447–3455 (2019).
- 718 28. W. Tan, X. Yao, T.-T. Le, *et al.*, "The application of optical coherence tomography angiography in systemic
719 hypertension: A meta-analysis," *Front. Med.* **8**, 778330 (2021).
- 720 29. X. Yao, M. Ke, Y. Ho, *et al.*, "Comparison of retinal vessel diameter measurements from swept-source OCT
721 angiography and adaptive optics ophthalmoscope," *Br. J. Ophthalmol.* **105**, 426–431 (2021).
- 722 30. H. Lee, M. Lee, H. Chung, and H. C. Kim, "Quantification of retinal vessel tortuosity in diabetic retinopathy using
723 optical coherence tomography angiography," *Retina* **38**, 976–985 (2018).
- 724 31. N. Mehta, K. Liu, A. Y. Alibhai, *et al.*, "Impact of binarization thresholding and brightness/contrast adjustment
725 methodology on optical coherence tomography angiography image quantification," *Am. J. Ophthalmol.* **205**, 54–65
726 (2019).
- 727 32. A. Rabiolo, F. Gelormini, R. Sacconi, *et al.*, "Comparison of methods to quantify macular and peripapillary vessel
728 density in optical coherence tomography angiography," *PLOS ONE* **13**, e0205773 (2018).
- 729 33. E. Borrelli, R. Sacconi, M. Parravano, *et al.*, "Optical coherence tomography angiography assessment of the diabetic
730 macula: A comparison study among different algorithms," *Retina* **41**, 1799–1808 (2021).
- 731 34. I. G. Freedman, E. Li, L. Hui, *et al.*, "The impact of image processing algorithms on optical coherence tomography
732 angiography metrics and study conclusions in diabetic retinopathy," *Transl. Vis. Sci. & Technol.* **11**, 7 (2022).
- 733 35. A. Arrigo, E. Aragona, A. Saladino, *et al.*, "The impact of different thresholds on optical coherence tomography
734 angiography images binarization and quantitative metrics," *Sci. Rep.* **11**, 14758 (2021).

36. J. Hong, B. Tan, N. D. Quang, *et al.*, "Intra-session repeatability of quantitative metrics using widefield optical coherence tomography angiography (OCTA) in elderly subjects," *Acta Ophthalmol.* **98**, e570–e578 (2020).
37. J. Hong, M. Ke, B. Tan, *et al.*, "Effect of vessel enhancement filters on the repeatability of measurements obtained from widefield swept-source optical coherence tomography angiography," *Sci. Rep.* **10**, 22179 (2020).
38. P. Cimalla, J. Walther, M. Mittasch, and E. Koch, "Shear flow-induced optical inhomogeneity of blood assessed in vivo and in vitro by spectral domain optical coherence tomography in the 1.3 μm wavelength range," *J. Biomed. Opt.* **16**, 116020 (2011).
39. J. Zhu, M. T. Bernucci, C. W. Merkle, and V. J. Srinivasan, "Visibility of microvessels in optical coherence tomography angiography depends on angular orientation," *J. Biophotonics* **13**, e202000090 (2020).
40. M. T. Bernucci, C. W. Merkle, and V. J. Srinivasan, "Investigation of artifacts in retinal and choroidal OCT angiography with a contrast agent," *Biomed. Opt. Express* **9**, 1020–1040 (2018).
41. Y. Pan, J. You, N. D. Volkow, *et al.*, "Ultrasensitive detection of 3D cerebral microvascular network dynamics in vivo," *NeuroImage* **103**, 492–501 (2014).
42. C. W. Merkle, C. Leahy, and V. J. Srinivasan, "Dynamic contrast optical coherence tomography images transit time and quantifies microvascular plasma volume and flow in the retina and choriocapillaris," *Biomed. Opt. Express* **7**, 4289–4312 (2016).
43. C. W. Merkle and V. J. Srinivasan, "Laminar microvascular transit time distribution in the mouse somatosensory cortex revealed by dynamic contrast optical coherence tomography," *NeuroImage* **125**, 350–362 (2016).
44. D. X. Hammer, R. D. Ferguson, T. E. Ustun, *et al.*, "Line-scanning laser ophthalmoscope," *J. Biomed. Opt.* **11**, 041126 (2006).
45. D. X. Hammer, R. D. Ferguson, A. H. Patel, *et al.*, "Angiography with a multifunctional line scanning ophthalmoscope," *J. Biomed. Opt.* **17**, 026008 (2012).
46. P. Garrone, G. Biondi-Zoccai, I. Salvetti, *et al.*, "Quantitative coronary angiography in the current era: Principles and applications," *J. Interv. Cardiol.* **22**, 527–536 (2009).
47. L. Zhou, M. S. Rzeszutarski, L. J. Singerman, and J. M. Chokreff, "The detection and quantification of retinopathy using digital angiograms," *IEEE Trans. Med. Imaging* **13**, 619–626 (1994).
48. J. Lowell, A. Hunter, D. Steel, *et al.*, "Measurement of retinal vessel widths from fundus images based on 2-D modeling," *IEEE Trans. Med. Imaging* **23**, 1196–1204 (2004).
49. C. Leahy, H. Radhakrishnan, G. Weiner, *et al.*, "Mapping the 3D connectivity of the rat inner retinal vascular network using OCT angiography," *Investig. Ophthalmol. & Vis. Sci.* **56**, 5785–5793 (2015).
50. G. Garhöfer, C. Zawinka, H. Resch, *et al.*, "Diffuse luminance flicker increases blood flow in major retinal arteries and veins," *Vis. Res.* **44**, 833–838 (2004).
51. M. Sharifzad, K. J. Witkowska, G. C. Aschinger, *et al.*, "Factors determining flicker-induced retinal vasodilation in healthy subjects," *Investig. Ophthalmol. & Vis. Sci.* **57**, 3306–3312 (2016).
52. G. Michelson, A. Patzelt, and J. Harazny, "Flickering light increases retinal blood flow," *Retina* **22**, 336–343 (2002).
53. G. C. Aschinger, L. Schmetterer, K. Fondi, *et al.*, "Effect of diffuse luminance flicker light stimulation on total retinal blood flow assessed with dual-beam bidirectional Doppler OCT," *Investig. Ophthalmol. & Vis. Sci.* **58**, 1167–1178 (2017).
54. T. E. Kornfield and E. A. Newman, "Regulation of blood flow in the retinal trilaminar vascular network," *J. Neurosci.* **34**, 11504–11513 (2014).
55. Y.-Y. I. Shih, L. Wang, B. H. De La Garza, *et al.*, "Quantitative retinal and choroidal blood flow during light, dark adaptation and flicker light stimulation in rats using fluorescent microspheres," *Curr. Eye Res.* **38**, 292–298 (2013).
56. T. Son, B. Wang, D. Thapa, *et al.*, "Optical coherence tomography angiography of stimulus evoked hemodynamic responses in individual retinal layers," *Biomed. Opt. Express* **7**, 3151–3162 (2016).
57. S. Fialová, M. Augustin, M. Glösmann, *et al.*, "Polarization properties of single layers in the posterior eyes of mice and rats investigated using high resolution polarization sensitive optical coherence tomography," *Biomed. Opt. Express* **7**, 1479–1495 (2016).
58. C. W. Merkle, M. Augustin, D. J. Harper, *et al.*, "High-resolution, depth-resolved vascular leakage measurements using contrast-enhanced, correlation-gated optical coherence tomography in mice," *Biomed. Opt. Express* **12**, 1774–1791 (2021).
59. Y. Geng, L. A. Schery, R. Sharma, *et al.*, "Optical properties of the mouse eye," *Biomed. Opt. Express* **2**, 717–738 (2011).
60. C. Schmucker and F. Schaeffel, "A paraxial schematic eye model for the growing C57BL/6 mouse," *Vis. Res.* **44**, 1857–1867 (2004).
61. D. J. Harper, M. Augustin, A. Lichtenegger, *et al.*, "Retinal analysis of a mouse model of Alzheimer's disease with multicontrast optical coherence tomography," *Neurophotonics* **7**, 015006 (2020).
62. Y. Fu and K.-W. Yau, "Phototransduction in mouse rods and cones," *Pflügers Arch. - Eur. J. Physiol.* **454**, 805–819 (2007).
63. I. A. S. Olsson, S. P. d. Silva, D. Townend, and P. Sandøe, "Protecting animals and enabling research in the European Union: An overview of development and implementation of Directive 2010/63/EU," *ILAR J.* **57**, 347–357 (2016).
64. M. Augustin, D. J. Harper, C. W. Merkle, *et al.*, "Optical coherence tomography findings in the retinas of SOD1 knockout mice," *Transl. Vis. Sci. & Technol.* **9**, 15 (2020).
65. M. Augustin, S. Fialová, T. Himmel, *et al.*, "Multi-functional OCT enables longitudinal study of retinal changes in a

798 VLDLR knockout mouse model,” PLOS ONE **11**, e0164419 (2016).

799 66. C. T. Sullender, L. M. Richards, F. He, *et al.*, “Dynamics of isoflurane-induced vasodilation and blood flow of cerebral
800 vasculature revealed by multi-exposure speckle imaging,” J. Neurosci. Methods **366**, 109434 (2022).

801 67. S. Worm, G. Ladurner, L. May, *et al.*, “Retinal OCT parameters and physiological dynamics as an effect of anesthesia
802 in C57BL/6J mice,” Investig. Ophthalmol. & Vis. Sci. **67**, 50 (2026).

803 68. M. Augustin, S. Fialová, R. Plasenzotti, *et al.*, “Multi-functional OCT image processing for rodent eyes,” in *Optical
804 Tomography and Spectroscopy*, (Optical Society of America, 2016), p. OTh4C.2.

805 69. T. T. Hormel, J. Wang, S. T. Bailey, *et al.*, “Maximum value projection produces better en face OCT angiograms than
806 mean value projection,” Biomed. Opt. Express **9**, 6412–6424 (2018).

807 70. G. Feng, A. Joseph, K. Dholakia, *et al.*, “High-resolution structural and functional retinal imaging in the awake
808 behaving mouse,” Commun. Biol. **6**, 572 (2023).

809 71. D. M. Stein, G. Wollstein, H. Ishikawa, *et al.*, “Effect of corneal drying on optical coherence tomography,”
810 Ophthalmology **113**, 985–991 (2006).

811 72. G. J. McLellan and C. A. Rasmussen, “Optical coherence tomography for the evaluation of retinal and optic nerve
812 morphology in animal subjects: Practical considerations,” Vet. Ophthalmol. **15**, 13–28 (2012).

813 73. A. F. Frangi, W. J. Niessen, K. L. Vincken, and M. A. Viergever, “Multiscale vessel enhancement filtering,” in *Medical
814 Image Computing and Computer-Assisted Intervention — MICCAI’98*, vol. 1496 of *Lecture Notes in Computer
815 Science* (Springer, Berlin, Heidelberg, 1998), pp. 130–137.

816 74. T. Son, G. Ma, and X. Yao, “Functional OCT reveals anisotropic changes of retinal flicker-evoked vasodilation,” Opt.
817 Lett. **49**, 2121–2124 (2024).

818 75. H. Radhakrishnan and V. J. Srinivasan, “Multiparametric optical coherence tomography imaging of the inner retinal
819 hemodynamic response to visual stimulation,” J. Biomed. Opt. **18**, 086010 (2013).

820 76. K. Polak, L. Schmetterer, and C. E. Riva, “Influence of flicker frequency on flicker-induced changes of retinal vessel
821 diameter,” Investig. Ophthalmol. & Vis. Sci. **43**, 2721–2726 (2002).

822 77. Z. Chu, J. Lin, C. Gao, *et al.*, “Quantitative assessment of the retinal microvasculature using optical coherence
823 tomography angiography,” J. Biomed. Opt. **21**, 066008 (2016).

824 78. W. Wei, Q. Zhang, S. G. Rayner, *et al.*, “Automatic vessel diameter quantification and vessel tracing for OCT
825 angiography,” J. Biophotonics **13**, e202000248 (2020).

826 79. S. A. Abbasi, D. Mei, Y. Wei, *et al.*, “Deconvolution techniques in optical coherence tomography: advancements,
827 challenges, and future prospects,” Laser & Photonics Rev. **19**, 2401394 (2025).

828 80. S. B. Ploner, E. M. Moulton, W. Choi, *et al.*, “Toward quantitative optical coherence tomography angiography:
829 Visualizing blood flow speeds in ocular: Pathology using variable interscan time analysis,” Retina **36**, S118–S126
830 (2016).

831 81. S. H. Yun, G. J. Tearney, J. F. de Boer, and B. E. Bouma, “Motion artifacts in optical coherence tomography with
832 frequency-domain ranging,” Opt. Express **12**, 2977–2998 (2004).

833 82. J. Walther, G. Mueller, H. Morawietz, and E. Koch, “Signal power decrease due to fringe washout as an extension of
834 the limited doppler flow measurement range in spectral domain optical coherence tomography,” J. Biomed. Opt. **15**,
835 041511 (2010).